

Wearable Augmented Reality Interface Design for Bridge Inspection

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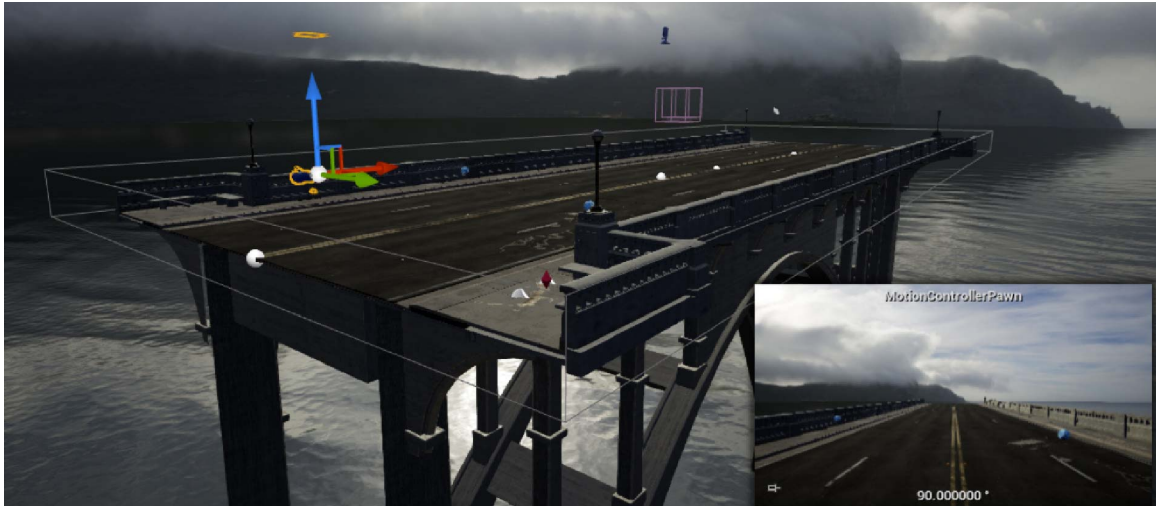


Figure 1: Prototype of augmented reality bridge inspection application made in Unreal Engine 4 and displayed in virtual reality

ABSTRACT

Condition assessment of high-consequence infrastructure, such as roadway bridges, involves substantial time and effort to ensure structural longevity and safety. Wearable augmented reality displays equipped with sensors and cutting edge computer vision techniques can supplement the inspection process, reducing mental fatigue and enhancing the inspection report workflow. Presented is the user experience process of designing a user interface for augmented reality to be used by bridge inspectors. Methods utilized for data collection and prototyping are discussed. A qualitative evaluation of the system revealed limitations and future steps in updated designs of the system.

Index Terms: Human-centered computing—Augmented Reality—Virtual Reality—Bridge Inspection; Computer Vision—User Experience—

1 INTRODUCTION

Bridges are integral components of public transportation infrastructure essential for facilitating the movement of goods and connecting communities. As such, routine on-site inspections are conducted by trained personnel every two years to ensure bridge longevity and public safety [1]. Condition assessment of bridges involves identifying material defects and monitoring potential problem areas to prevent major structural issues from developing. Bridge inspections are therefore methodical tasks demanding substantial time, attention to detail, and precise documentation.

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Due to bridge inspection being largely vision-based, development of computer-vision (CV) techniques is an ongoing field of interest as a means to facilitating visual search tasks, documentation of structural health, and three-dimensional scene reconstruction [23]. Currently, the use of unmanned aerial vehicles (UAVs) is a leading platform supporting CV research for infrastructure inspection [11, 14]. While UAVs reduce risk imposed on workers and afford state transportation agencies a means to accessing hard-to-reach areas, they currently require specialized training and license. As such, we posit that augmented reality (AR) devices equipped with visible light cameras, depth cameras, and infrared sensors can be incorporated into the assessment of high-consequence infrastructure. Additionally, CV methods developed for UAV inspection applications can be applied to wearable AR displays equipped with similar sensing devices to supplement boots-on-the-ground inspection [9, 10].

AR-based CV methods afford bridge inspectors less ambiguous visual inspections by reducing subjectivity placed in measurements and enhancing the inspection reports with quantitative data. Image analysis algorithms have been developed and modified for material segregation, component and element identification, and defect detection [2, 22, 24, 25]. Bianchi et al. used CV machine learning models to detect defects and contextualize them by associated structural details [3]. Recent adjustments to inspection report requirements made by the Federal Highway Administration seek to improve the quality of element-level inspection data, further increasing the need for sophisticated CV solutions [20].

The use of wearable AR for vigilance tasks such as defect detection and annotation will accelerate the overall inspection process while reducing worker fatigue. Hammad et al. provided an initial discussion on the challenges of employing AR applications in the bridge inspection domain, covering human-computer interaction methods and a concept for an early prototype [12]. Chettupuzha et al. demonstrated the feasibility of utilizing virtual reality (VR) to evaluate an AR bridge inspection interface prototype, identifying necessary interface considerations [7]. A collaboration of researchers from

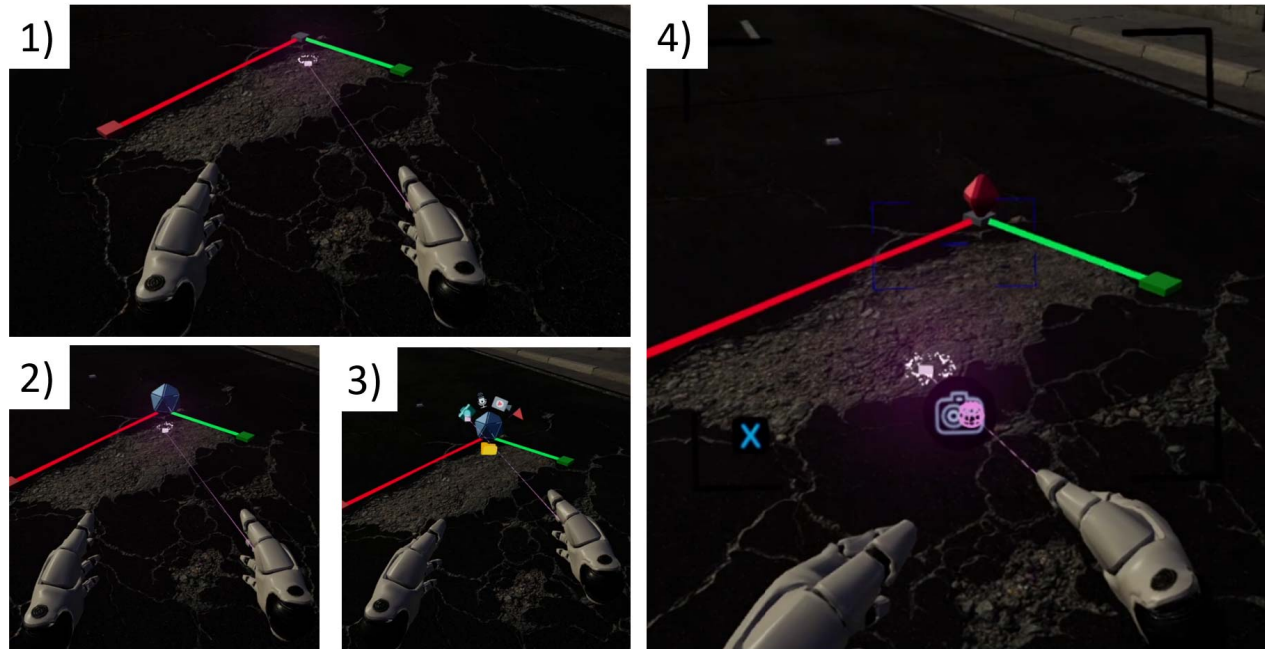


Figure 2: Defect orb spawning and adding data 1) point gesture 2) click to spawn orb 3) click to open radial menu and point/click to select camera 4) take picture with camera HUD which adds image data to the container

University of New Mexico and Los Alamos National Laboratory introduced a technical framework for implementing AR into the bridge inspection process [17]. The team developed various AR applications to collect and assess the accuracy of surface area measurements, surface crack measurements, and depth measurements [15, 16, 26].

Along with data collection methods, Nguyen and Shim et al. presented a working prototype of an AR system that merges bridge inspection data collected in AR with building information modeling (BIM) techniques [8]. The resulting model can be imported into bridge management systems and updated via a layered approach to documentation and historical tracking. This method of data management offers new digital visualization methods for large data sets that can be viewed and evaluated collaboratively and remotely [18, 21].

While the aforementioned studies show progress made in technical developments of computer-assisted infrastructure inspection, there remains a need for research into the design of user interfaces that promote a positive user experience (UX). The following details the initial steps taken to design a wearable AR interface for on-site bridge inspection report generation with a focus on UX.

2 PROJECT OVERVIEW

To determine the best way to integrate algorithms such as crack detection into existing inspection workflows, the authors are working to develop and test an AR platform in collaboration with the Virginia Department of Transportation (VDOT). The long term desired features of the project intend to support (in order of increasing complexity): 1) Geo-located Annotations, 2) Automated/Semi-Automated Defect Labeling, 3) Structural Context Awareness, and, 4) Interactive 3D Digital Twins. These features collectively will assist the inspector in the field as well as structuring the resultant data in such a way as to allow for automated report generation after inspection. This paper specifically focuses on developing a prototype AR framework that can address features 1 and 2 of the above, while providing a scaffold for which features 3 and 4 can be built upon in future work.

3 DATA COLLECTION

Prior to the creation of the AR system, usage research data was collected to better understand the work domain. The authors met with professionals at various levels of the bridge division organization structure to achieve a high-level view of various user roles that would be interfacing with the AR system. Contextual inquiries were made via person-to-person interviews, surveys, and on-site bridge inspection shadowings.

The interviews consisted of 20 virtual meetings with various Department of Transportation personnel across the country. Two separate online surveys were sent out, one to VDOT bridge inspectors (those doing the inspections), and one to bridge inspector engineers (those managing the inspections). The latter received six responses, but this target audience overlapped significantly with those of the interviews. The former received 41 responses, which consisted of unique viewpoints from less senior personnel than those of the interviews. Last, on-site research was conducted in which the authors observed real-time inspections of three bridges and one culvert.

The data gathered from interviews and surveys allowed for the development of a coherent flow model of the work domain, detailing relationships between system actors and work domain artifacts. Subsequently, a hierarchical task analysis was conducted to distill the data into a comprehensive task flow structure and identify the various steps involved in the bridge inspection process. Inspectors will begin pre-inspection by reviewing and printing the previous inspection report prior to traveling to the site. This paper document is roughly 20 pages in length, and consists primarily of narrative text detailing each inspection note related to defects or potential problem areas. Inspection teams usually consist of two persons, one of which reads the previous report and directs the other inspector to each location. The inspector at the defect location annotates and measures physical defects or problem areas and relays the current status back to the first inspector to mark changes on the report. After the inspection, these hand written notes and pictures are updated in the digital inspection report document which is manually input

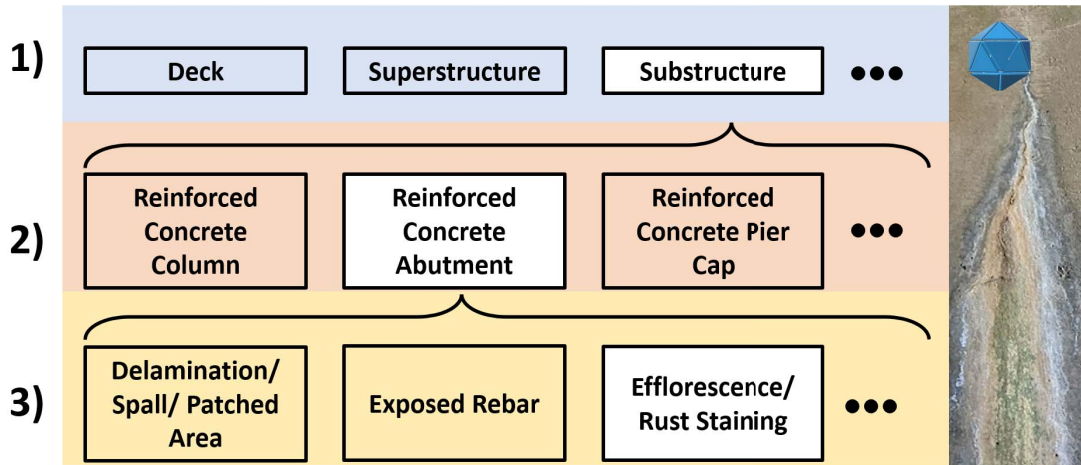


Figure 3: An example window progression to flag an efflorescence defect on a reinforced concrete abutment by stepping through the 1) bridge section 2) bridge element and 3) defect to be documented

into multiple separate systems, including load rating software and American Association of State Highway and Transportation Officials (AASHTO) bridge management software (BrM), to be used by various other departments.

Barriers identified throughout the data elicitation phase were mainly a consequence of the outdated narrative note-taking method. By lacking a standardized base system to organize the information, the data must be manually entered into multiple separate systems. It was regularly mentioned in interviews that handwriting legibility issues and repeated entry of information led to user mistakes and quality control problems. It was also noted from the interviews that defects can often be difficult to measure using a handheld tape measure, leading to subjectivity in how these measurements are approximated. Along with this, it was observed during on-site visits that finding a defect location with only narrative textual descriptors can sometimes be challenging. It should also be mentioned that a barrier distilled from inspector surveys revealed that while there is interest in the use of innovative devices, there is a general mistrust of technology implementation as a result of failed adoptions of devices such as tablets and UAVs in the past.

4 PROTOTYPE CREATION

To address these barriers an AR inspection assistance tool was envisioned, referred to as the Wearable Inspection Report Management System (WIRMS) hereafter. The system was designed to provide the ability for inspectors to identify and mark problem areas on the bridge during inspection, reflecting the existing workflow of annotating defects with a marker or chalk. The system treats these marked locations, or flags, as folder structures anchored in 3D space, containing metadata such as recorded notes, digital measurements and annotations, and location information corresponding to the defect. Storing metadata associated with each defect could allow for inspection reports to be automatically generated and distributed, reducing time spent digitizing reports post-inspection and providing a more cohesive approach to report management.

Given the difficulty of iteratively fielding AR applications in ad hoc outdoor locations, we decided to focus on first making a virtual reality (VR) prototype of the AR WIRMS experience to allow for more rapid iteration and evaluation of the user interface design. This VR prototype was developed in Unreal Engine 4 and tested on the Oculus Quest 2 (Figure 1). Subsequent and final versions of the system will be developed in Unity and deployed on head-worn AR displays such as the Microsoft HoloLens 2.

In the VR-based prototype, the intended AR gestures (tap and pinch) were replaced with motion controller input actions (point and click) for scene interaction. Upon selecting a problem area, WIRMS places a virtual 3D orb object which resembles a container or folder metaphor to hold information related to the defect (Figure 2). If the orb is selected, a radial menu appears providing options for data entry within the orb. The color and shape of the orb are intended to convey to the user information about the defect. As shown in Figure 2, the shape of a defect switches from a blue orb to a red diamond after an image has been taken of the defect. Similarly, an existing defect from a previous inspection could be colored or shaped differently to convey to the inspector that the area requires review.

Upon selecting the radial menu option indicated by the red flag icon, the user is prompted to step through a three menu dialog to identify the structural component and associated defect to generate a report of the defect (Figure 3). The WIRMS system requires the user to manually input this information following the AASHTO hierarchy of bridge elements [19]. In this way, users are: 1) directed to identify the section of the bridge first (superstructure, substructure, etc.), 2) presented with possible elements for that section of the bridge, and lastly, 3) prompted with the list of possible structural defects associated with the chosen bridge element. This type of organization is already present in the current narrative inspection reports and should be familiar to the intended users of WIRMS. However, in future implementations of the system, structural context awareness algorithms will be utilized to remove the need for manual entry. For example, a set of machine learning models could be used to identify the presence of a steel bearing with corrosion, thus bypassing the menu sequence previously described by automatically selecting the appropriate element and defect information.

After identifying the component and defect type in the aforementioned three step menu, the user would then quantify defect severity. In future instances of WIRMS, the component/defect information could be used to determine which CV technique to apply for defect quantification, such as corrosion state assessment for a steel girder or crack detection in concrete. These algorithms would provide measurements such as crack length and width, which could be used as a second opinion to user generated measurements. Currently, in the absence of computer automated measurement, users are able to create a virtual ruler object to annotate and measure various properties of the defect as seen in Figure 2. The advantage of these virtual measurements is that they can remain permanently affixed to the defect

container for future review. Furthermore, geolocation data associated with defect containers and associated digital annotations would enable accurate placement of defect markers in a reconstructed 3D bridge models for improved data visualization. This could be utilized by inspectors in the office for an off-line digital walk-through prior to arriving on site.

4.1 WIRMS Prototype Evaluation

The VR WIRMS prototype was evaluated in three stages to receive feedback on system usability. The first and second stages consisted of a single user pilot study and an inter-team system evaluation respectively. The third stage consisted of a "Rapid Empirical Evaluation" with four Subject Matter Experts (SMEs). While we were unable to access VDOT personnel in person due to COVID restrictions, we were able to recruit a Civil Engineering professor, graduate student, and two undergraduate students to stand in as SMEs.

These evaluations were intended to gauge how well the mental model of the designers matched that of the prospective users. In the first two stages of evaluation, this was primarily focused on minor bug fixes to allow for a more productive evaluation in the third stage.

Key takeaways from these evaluations can be divided into issues with the current system and suggestions for additions onto the system. The largest issue found with the current system was that the AR menus associated with defect creation were presented in a head-referenced fashion (i.e., head-locked as with traditional HUDs), and users found them frustrating to interact with. This could be mitigated by either pinning these menus to the location at which they were spawned (i.e., a world-referenced graphic), or simply adding a delayed-follow to the head-tracked movement (i.e., a tethered body-relative graphic). Another issue was that the logic for the radial menu was setup to prevent another orb from being selected unless the previous orb's radial menu was closed. Participants would forget to close the menu, thus requiring them to redirect focus to the previous defect. This logic was changed so that each new selected defect closes the previous radial menu if not done so already.

A common action during the inspection process is looking at prior marked defects and checking for signs of deterioration. The current interface did not provide the ability to confirm if existing data was accurate. Similarly, a typical inspection involves checking a sequential list of items. Users expressed interest in viewing a menu containing all the existing defect orbs, possibly organized by the AASHTO hierarchy as their existing narrative inspection report is formatted. Lastly, it was mentioned that integration of vocal commands would be helpful for both menu selection and voice-to-text note taking.

5 DISCUSSION & FUTURE WORK

The prototype developed in this study provides a design framework for an AR bridge inspection and reporting system. The main issues left to be resolved are decisions of which data analysis techniques should be incorporated, and how to best manage environment mesh localization and long term stabilization.

As for the former, the authors of this study have been working with Bianchi et al. in order to incorporate various machine learning models [3]. These include material segmentation in image data, steel corrosion condition state assessment, and concrete crack detection [4–6]. The HoloLens is equipped with various camera sensors as well as a depth sensor. By applying image processing techniques to the images captured by the camera sensors, we can overlay the results of these algorithms and segment, or classify, the corresponding 3D points captured by the depth sensor. This will allow image based techniques to accurately interface with the HoloLens' 3D environment spatial mesh.

Material segmentation could be applied to pre-filter the list of possible components that can be selected when the user flags a particular defect. Similarly, corrosion state and crack detection

algorithms could be incorporated into an automated annotation and measurement tool available to defects associated with steel and concrete respectively. The aim of developing the WIRMS prototype and user interface assessment was to provide a framework for which these technologies could be implemented to test in future field-based studies.

A separate issue that was discovered when working with the HoloLens in bridge inspection environments is that the built-in spatial mesh generation process is not globally stable, nor geo-referenced to an absolute world location. As the user looks around, new data is incorporated and the surrounding mesh is continuously making minor adjustments to ensure that the immediate surroundings are as accurate as possible. While this is efficient for indoor environments and optimizing near-field hologram stability, it is not adequate for anchoring annotations which must remain in their location for years without moving. Additional hardware and supporting software will need to be integrated and developed to mitigate this issue, and may include incorporating DGPS and LiDAR as shown in Hansen et al. [13]. Another possible solution would involve leveraging a "digital twin" as presented by Nguyen and Shim et al. to overlay a static 3D model of the bridge to which items could be anchored to, while utilizing the built in mesh to fine-tune the positioning of the model [8].

6 SUMMARY

This paper intends to share the authors' understanding of the bridge inspection work domain, as derived from a UX process, and offers further insight into how a prospective AR system might integrate into the inspection workflow. While there have been a large number of studies developing novel methods of automating inspection tasks, few focus on how they might actually be incorporated into the work itself. Those studies which do consider this task often utilize UAVs, which have unique use cases and limitations.

AR based systems have the potential to integrate seamlessly into the existing manual inspection workflow while still providing a platform through which inspection assistance research can be incorporated. This particular unification could reduce mental workload by providing an annotated virtual overlay of their surroundings. Rather than reading through paragraphs of directions to find a specific defect, inspectors could simply look around their environment and see defect position highlighted in 3D space. The system offers reduced subjectivity in user measurements by incorporating cutting edge image processing algorithms to automatically measure defects such as cracks. By migrating the data collection process to a digital system, a single database could be developed to interface with all of the other systems automatically. This could mitigate quality control issues associated with manual user input. Ultimately, the use of WIRMS would allow inspectors to focus more on safety-critical tasks such as identifying defect locations while simultaneously assisting in the inspection reporting process.

As the field of CV and machine learning based inspection algorithms matures, more focus is needed to determine how these tools will be incorporated into industry. The future of infrastructure maintenance will likely see a combination of tools including AR and drone-based imaging, therefore it is important to consider the role of the human operator. Rather than replacing inspectors currently performing these safety-critical tasks, a human-centric approach ensures trained engineers remain a crucial part of the process.

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